

Overview of published files – BIRD 6.7 and Transformation Rules 3.1

This document provides additional details regarding the BIRD Data Models 6.7 and Transformation Rules 3.1 publication described on the [BIRD website](#). A special focus is given to how the components are connected to each other. Please be aware that all documents and material are published “as is” reflecting the current state of development of the BIRD Logical Data Model, BIRD Transformation Rules, and the accompanying documentation. Enhancements are displayed in a transparent manner by highlighting the most important changes from one version to another. Additional information regarding the BIRD process and components can be found on the [BIRD methodology](#) page.

1. Logical Data Model/Enriched Logical Data Model

Logical Data Model (LDM)

The BIRD LDM is a highly normalised data model describing the data necessary to fulfil the reporting requirements covered by the BIRD. Its purpose is the logical description of the data (“what needs to be reported”). This means that dependencies and therefore implicit validations rules are covered within the model at very detailed level.

Enriched Logical Data Model (ELDM)

The ELDM is similar to the LDM but includes additional derived information that is needed to fulfil the regulatory reporting requirements. For example, the enriched concept of small and medium enterprise classification is derived from primary concepts contained in the LDM, such as the number of employees and annual turnover.

The LDM/ELDM are implementation agnostic and can be used by banks and other interested parties to design their own physical implementation layers.

To demonstrate the applicability, BIRD provides a physical implementation model based on the LDM/ELDM which are the Input Layer (IL) & Enriched Input Layer (EIL), explained in the next section.

Published BIRD LDM documentation

- **Introduction to the LDM** - A detailed introduction to the LDM is available describing in a user-friendly way the content of the LDM/ELDM including additional explanatory material intended for business users.
- **BIRD LDM design principles** - The BIRD LDM is designed based on clear underlying design principles which are described in a separate document.
→ It is recommended that both documents but especially the "Introduction to the LDM" are studied before investigating the LDM/ELDM in detail.
- **Overview of changes to the BIRD content** – A detailed list of changes to the BIRD Logical Data Model, BIRD Transformation Rules, and the accompanying documentation.
- **BIRD metadata manual** – Comprehensive guide for BIRD users on how to access, navigate, and integrate BIRD metadata, including metadata related to the BIRD Logical Data Model.

Published BIRD LDM content

- The **LDM/ELDM** itself is published via SQL data modeler format. A publication based on the SMCube information model via an access database will follow soon.
→ Please be aware that the LDM/ELDM are published together comprised in the same data model called, for simplification purposes, LDM. The enriched parts are however clearly distinguishable as described in the "Introduction to the LDM" document.

Status

The LDM/ELDM are already very advanced and will be continuously improved and enhanced. There will be targeted and widespread review taking place in the BIRD community in preparation of the envisaged collaboration between the authorities and the banking industry regarding the Integrated Reporting Framework (IReF).

2. Input Layer/Enriched Input Layer

Input Layer (IL)

The IL is a less normalised¹ and forward engineered Entity Relationship Model (ERM) fully based on the LDM. This characteristic makes it a good candidate to be used as an interface for banks' internal IT systems and it could be the starting point for a technical implementation of BIRD. Moreover, the less normalised structure could allow for an easier understanding of the content for BIRD users that are not too familiar with complex data modelling applications.

Enriched Input Layer (EIL)

The EIL is a wrapped-up and forward engineered Entity Relationship Model (ERM) fully based on the ELDM and includes the enriched information. The same considerations as for the IL hold true on its use as an implementation layer or for easy understanding of the content by BIRD users.

¹ "Wrapped-up" here means that the information in the subtypes from the LDM are rolled-up into their ultimate supertype and are represented in one table in the input layer for implementation purposes.

The enriched information is in the EIL also identifiable as attributes/variables do have the prefix "Derived_" or "DRVD_".

The IL/EIL represent a potential physical implementation of the BIRD providing a basis for a potential standardised interface across banks to fulfil reporting requirements. Banks are advised to add their own solutions for various implementation concerns – like historization – that are not covered by BIRD.

Published BIRD IL documentation

- **BIRD metadata manual** – Comprehensive guide for BIRD users on how to access, navigate, and integrate BIRD metadata, including metadata related to the BIRD Input Layer.
- **Forward engineering (FE) in the context of BIRD** - Description of the forward engineering methods applied to the LDM/ELDM to create the IL/EIL, and their implications. The described methods allow for an automated creation of the IL/EIL and require limited maintenance effort.
- There is not separate document available describing the IL/EIL. Considering that the IL/EIL are fully forward engineered from the LDM/ELDM the explanatory material available for these BIRD components in conjunction with the explained forward engineering procedure should provide a sound basis for the understanding of the IL/EIL.

→ *It is recommended that the document regarding the forward engineering procedure is studied before investigating the IL/EIL in detail.*

→ *Please be aware that there are two publications, one being the IL only and another being the EIL including the enriched components. The enriched (i.e., derived) parts are clearly distinguishable as described in the "Forward engineering in the context of BIRD" document.*

Published BIRD IL content

- The **IL/EIL** itself is published via SQL data modeler format. A [Restful API](#) is also available from the website.
- **Forward engineering (FE) scripts** are published as a part of SQL data modeler files inside a folder called *forward engineering*. The xml file contains the forward engineering scripts, which can be imported into SQL data modeler and used as instructed in the *fe_instructions.txt* file.
- **Validation rules for the IL/EIL** – Two json files containing structural validation rules for the IL/EIL, one contains constraints on the relationships or foreign keys and the second contains constraints on the columns of the IL/EIL.
- A publication based on the **SMCube information model** is also available, via the "[Metadata & Exports](#)" link at the top right of the BIRD webpage.

Status

The IL/EIL represent what can be derived from the LDM/ELDM. The BIRD community will evaluate, improve, and enhance the existing version via a targeted and widespread review in preparation of the envisaged collaboration between the authorities and the banking industry regarding the Integrated Reporting Framework (IReF).

Please be also aware that for the IL/EIL to be fully equivalent to the LDM/ELDM in terms of content does require validation rules. These can also be automatically generated via the FE scripts. The format validation rules included represents the first version of the validation rules to be published.

3. Transformation Rules

Transformation Rules

Within BIRD, there are two types of logical/semantic transformation rules: derivation and generation rules. Both are used to explain how data is transformed from the Input Layer to produce the Output Layer (i.e., specific reporting templates/tables).

Derivation rules describe how input data found in the BIRD Data Models can be used to calculate derived information. For example, the calculation of the gross carrying amount. Generation rules describe how a specific reporting template/table can be generated starting from the BIRD Data Models. The derivation phase (i.e., enrichment of concepts using primary concepts available in the input) starts from the Logical Data Model (LDM) or the Input Layer (IL). The generation phase (i.e., creation of output reports by filtering, aggregating, and selecting existing concepts) takes place from the Enriched LDM (ELDM) or the Enriched IL (EIL).

Published BIRD Transformation Rules documentation

- **Overview of changes to the BIRD content** – A detailed list of changes to the BIRD Logical Data Model, BIRD Transformation Rules, and the accompanying documentation.
- **BIRD metadata manual** – Comprehensive guide for BIRD users on how to access, navigate, and integrate BIRD metadata, including metadata related to the BIRD Transformation Rules.

Published BIRD Transformation Rules content

Please note that the Excel files with the compilations of Derivation Rules and Generation Rules have been discontinued in April and November 2025, respectively. A complete list of Derivation and Generation rules can be found via the Framework logical lineage export ([Metadata & Export section](#)).

Status

Publication of FINREP (77 out of 122 templates), Asset Encumbrance, and AnaCredit Transformation Rules, using BIRD Data Models version 6.6.

For further information or in case of questions, contact the European Central Bank's BIRD team at BIRD@ecb.europa.eu